DOD-STD-1312-105
26 March 1984

SUPERSEDING
DOD-STD-1312-5
19 October 1983

MILITARY STANDARD

FASTENER TEST METHODS, METRIC

METHOD 105

STRESS DURABILITY

FSC 53GP

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE
WASHINGTON, DC 20301

Fastener Test Methods, Metric, Method 105, Stress Durability

DOD-STD-1312-105

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Code 931, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this standard or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

FOREWORD

This standard sets forth standard test procedures for determining the capability of externally threaded fasteners to withstand the various stress load conditions encountered when used in military weapons systems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

TABLE OF CONTENTS

Paragraph		Page
1.	SCOPE	1
1.1	Applicability	1
2.	REFERENCED DOCUMENTS	1
2.1	Government documents	1
2.1.1	Specifications, standards and handbooks	1
3.	DEFINITIONS	1
4.	GENERAL REQUIREMENTS	1
4.1	Test apparatus	1
4.1.1	Torque method of loading	1
4.1.2	Elongation method of loading	2
4.1.3	External loading method	2
4.1.4	Load cell method of loading	2
4.2	Test specimen sample size	2
5.	DETAIL REQUIREMENTS	2
5.1	Test methods	2
5.1.1	Torque method	2
5.2	Elongation method	3
5.3	External loading method	3
5.4	Load cell method	4
5.6	Specimen examination	4
5.7	Test results	4
6.	NOTES	4
6.1	Test reports	4

FIGURES

Figure	Title	Page
1.	Test block	5
2.	Test spacer	5
3.	Tapped block	6
4.	Torque and elongation testing assembly	6
5.	Testing assembly	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Applicability. This method establishes a standard procedure for testing the stress durability of externally threaded fasteners. This procedure applies to all types of externally threaded fasteners which may be subject to any type of embrittlement and is not limited by configuration or size.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the current Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto (if applicable), form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

GGG-W-686	Wrench, Torque

MILITARY

MIL-T-5544	Thread Compound, Antiseize, Graphite-Petrolatum
MIL-I-6866	Inspection; Penetrant Method of
MIL-I-6868	Inspection Process, Magnetic Particle

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. DEFINITIONS

Not applicable.

4. GENERAL REQUIREMENTS

4.1 Test apparatus.

4.1.1 Torque method of loading. Torque wrenches calibrated for accuracy within the limits specified in GGG-W-686 will be required, as well as the necessary adaptors to fit the configuration of the fastener under test. Also required are steel blocks as shown on figure 1, heat treated to a minimum hardness of Rockwell C 45, or steel spacers as shown on figure 2, heat treated to a minimum hardness of Rockwell C 45, and an internally threaded member (figures 3 and 4) with sufficient strength to develop full tensile strength of the fastener being tested.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

4.1.2 Elongation method of loading. Equipment in accordance with 4.1.1 shall be used, except that any suitable wrenching device may be used and a precision measuring instrument is required to measure elongation of the test fastener according to acceptable methods.

4.1.3 External loading method. Suitable equipment and fixtures such as those used in tensile testing of fasteners shall be used to develop and maintain the required test load on the fastener.

4.1.4 Load cell method of loading. A precision calibrated load cell of sufficient strength to develop full tensile strength of the fastener under test shall be used, and be of proper dimensions to accommodate the fastener. Strain gages shall be fastened by standard approved methods of the load cell as indicated on figure 5. Hardened steel washers or plates as shown on figure 5 of sufficient strength to develop full tensile strength of fastener shall be required. In addition, an accurate strain indicator is required which, at the option of the tester, may be coupled with a multi-channel switch and balance unit. Also required are a suitable wrenching device for the test fastener and an internally threaded member as specified in 4.1.1.

4.2 Test specimen sample size. The number of specimens to be tested from each lot shall be as specified in the acquisition document or product specification. The test specimen shall not be altered other than as specified in 5.2.

5. DETAIL REQUIREMENTS

5.1 Test procedures. Unless otherwise specified, stress durability tests shall be conducted in accordance with one of the following methods listed below. Selection of method shall be an option of the manufacturer. In the event of controversy, the external loading method shall be used as the referee method.

5.1.1 Torque method. Test fastener shall be assembled in the test units, utilizing not more than three spacers in accordance with figure 4. The threads of the fastener, the bearing surface of the fastener head, and the bearing surface of the nut shall be lubricated with a lubricant conforming to MIL-T-5544. A minimum of two full threads of the test specimen shall extend above the face of the nut or tapped block toward the head of the bolt. The tapped test block or the untapped block nearest to the nuts shall be held in a vise. The specimen shall be tightened with a torque wrench to a torque equivalent to the induced load. Unless otherwise specified, the induced load shall be 75-80 percent of the minimum ultimate tensile strength specified in the product specification. The specimen assembly shall remain in a torqued condition at room temperature for a period of time specified by the procurement document.

5.1.1.1 The torque value may be determined from the following equation or torque tension curves developed from tests utilizing hardened steel threaded members:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

$$T = KDL$$

Where: T = torque in Newton-meter
K = 0.1 (coefficient of friction)
D = nominal diameter of specimen (mm)
L = induced load (kN)

5.2 Elongation method. The test fastener shall be assembled in the test units, utilizing not more than three blocks or spacers in accordance with figure 4. The threads of the fastener, the bearing surface of the fastener head, and the bearing surface of the nut shall be lubricated with a lubricant conforming to MIL-T-5544. A minimum of two full threads of the test specimen shall extend above the face of the nut or tapped block toward the head of the bolt. The tapped test block or the untapped block nearest the nut shall be held in a vise. The specimen shall be tightened with any suitable wrenching device until the predetermined elongation of the specimen has been reached by actual measurement with a precision measuring device, usually a micrometer, over the ends of the specimen. The assembly shall be allowed to stand at room temperature for a period of time specified by the procurement document. The predetermined amount of elongation shall be obtained from a load versus elongation curve developed from data obtained utilizing a standard tensile testing machine, comparable fixtures, and fasteners from the lot being evaluated. The test load shall be equal to 75-80 percent of the minimum ultimate tensile strength of the detail specification. The predetermined elongation can be computed using the formula:

$$e = \frac{P}{E}\left(\frac{L}{A} + \frac{L_t}{A_t}\right)$$

Where: e = total bolt elongation (mm)
P = Clamping force (N)
L = Length of grip (mm)
A_t = Thread stress area (mm^2)
L_t = Thread length between bearing faces (mm)
E^t = Modulus of elasticity (MPa)
A = Area of shank (mm^2)

NOTE: Approximately two pitches should be added to the thread length portion when using a nut.

If the condition of the end surfaces on which the measurement is taken is not suitable for precise measurement, the surfaces shall be prepared by grinding or center-drilling before assembly in the test fixture.

5.3 External loading method. The specimen shall be assembled in test fixtures designed to assure axial loading in static load type testing equipment. Sufficient weight shall be applied to develop an induced load of 75-80 percent of the minimum ultimate tensile strength of the product specification. A minimum of two full threads of the test specimen shall extend above the face of the nut or tapped block

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

toward the head of the bolt. The specimen shall remain in this condition at room temperature for the period of time specified by the procurement document.

5.4 Load cell method. The test specimen shall be assembled with the washers, the calibrated load cell to which strain gages have been attached, and the nut as shown on figure 5. The assembly shall be lubricated as shown on figure 5. The remainder of the procedure shall be performed in accordance with 5.1.1 except that the strain indicator with leads attached to the strain gages shall be used to indicate when the proper load has been reached and is being maintained throughout the test period.

5.5 Any other method of determining load which results in a load of 75-80 percent of the product minimum ultimate tensile strength being induced in the specimen is acceptable.

5.6 Specimen examination. The test specimen shall be disassembled and inspected for cracks in accordance with MIL-I-6868 for magnetic materials or MIL-I-6866 for other materials.

5.7 Test results. Cracks or fracture of any test specimen constitutes failure of the specimen.

6. NOTES

6.1 Test reports. The test reports shall contain the following data:

a. Fastener description.

b. Part number.

c. Lot identification.

d. Manufacturer.

e. Material.

f. Fastener strength level.

g. Installation torque or elongation.

h. Stress durability test load.

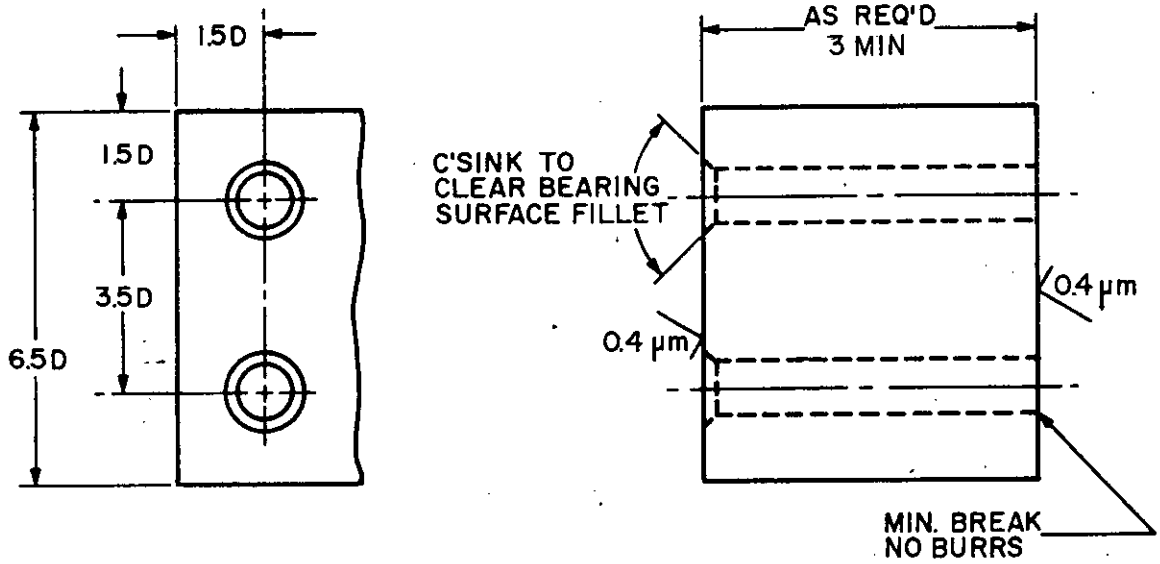
i. Test results.

j. Description of test method.

k. Calculations.

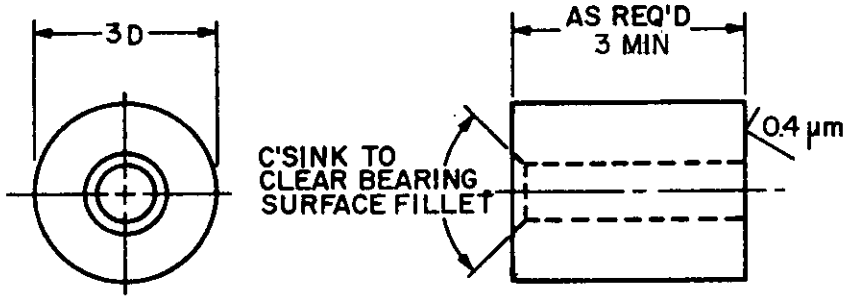
l. Test time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.



NOTES:
1. 1 or more holes perpendicular to top and bottom within $\pm 0^{\circ}$ 30'.
2. Diameter to fit shank of fastener as close as possible without interference.
3. D = Nominal diameter of fastener.
4. Top and bottom parallel within 1°.
5. Countersink for flush head fastener.
6. Dimensions in mm.

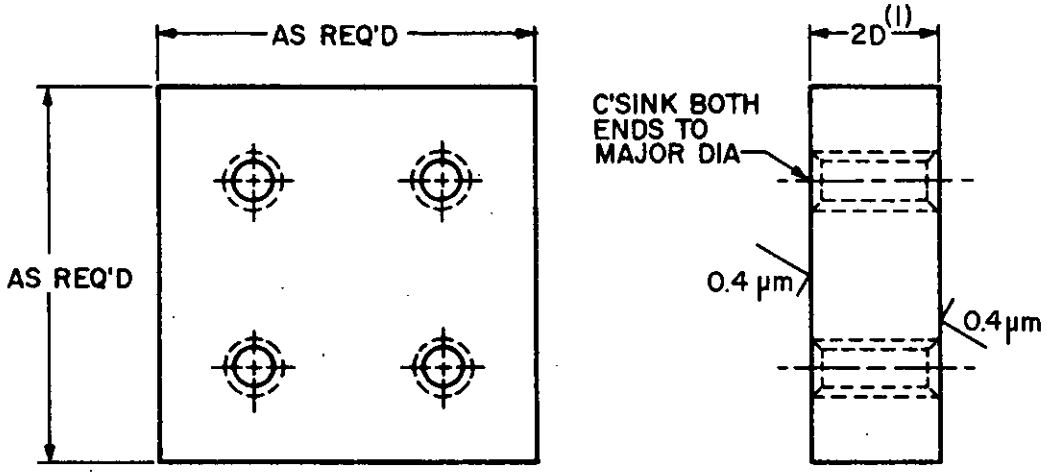
FIGURE 1. Test block.



NOTES:
1. Hole perpendicular to top and bottom within 0° 30'.
2. Top and bottom surfaces parallel within 1°.
3. D = Nominal diameter of fastener.
4. Diameter to fit shank of fastener as close as possible without interference.
5. Countersink for flush head fastener.
6. Dimension in mm.

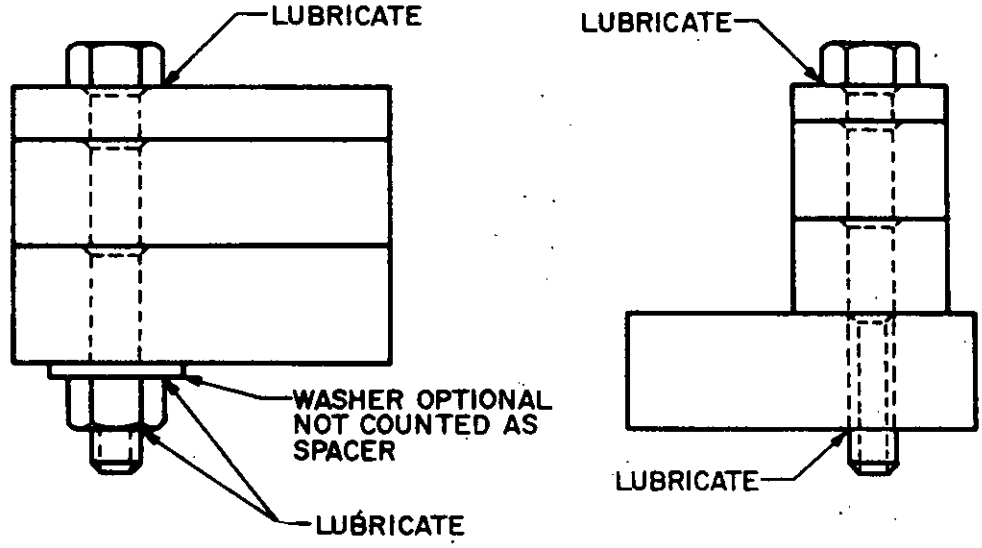
FIGURE 2. Test spacer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.



NOTES:
1. D = Nominal fastener diameter.
2. Thread tolerance 4H6H to MJ5 and 4H5H for larger sizes.
3. Top and bottom surfaces parallel within 1°.
4. Holes located to provide clearance for spacer and match figure 1 block.
5. Hole perpendicular to top and bottom within 0° 30'.

FIGURE 3. Tapped block.



NOTE:
1. Hole size D $+.02\ ^{+0.11}_{-0}$ within 1D of ends.
2. Internally threaded member must not contact the bearing face of the block or spacer having chamfered holes.
3. Dimensions in mm.

FIGURE 4. Torque and elongation testing assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

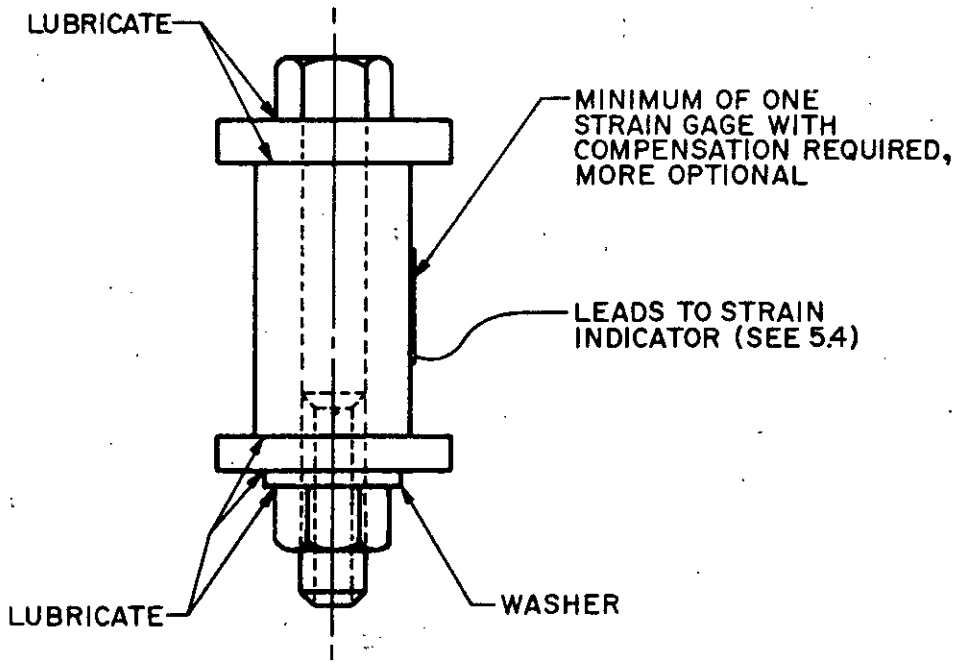


FIGURE 5. Testing assembly.

Custodians:
Army - AR
Navy - AS
Air Force - 11

Reviewer activities:
Army - AV, AR
Navy - AS, SH
Air Force - 11
DLA - IS

Preparing activity:
Navy - AS
(Project 53GP-0142-14)

☆ U.S. GOVERNMENT PRINTING OFFICE: 1984-705-040/A2097

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Engineering Specifications & Standards
Department (ESSD) Code 93
Lakehurst, NJ 08733

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.